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Thm. Let  $K$  be a field, and  $K_1, K_2 \leq K$  be subfields.

Define a composite field of  $K_1$  and  $K_2$  as:

$$K_1 K_2 \stackrel{\text{def}}{=} \bigcap_{\substack{F \leq K \\ K_1, K_2 \leq F}} F$$

That is, the smallest subfield of  $K$  containing  $K_1 \cup K_2$ .

Lemmu. Let  $K/F$  be a extension field, and let  $\alpha_1, \alpha_2 \in K$ . Then,

$$F(\alpha_1)F(\alpha_2) = F(\alpha_1, \alpha_2)$$

Proof. Since  $F(\alpha_1, \alpha_2)$  contains both  $F(\alpha_1)$  and  $F(\alpha_2)$ ,  $\subseteq$ , and since  $\alpha_1, \alpha_2 \in F(\alpha_1)F(\alpha_2)$ , so proved.

Thm. Let  $K/F$  be an Extension, and  $K_1, K_2 \leq K$  be finite extensions for  $F$ . Then,

$$[K_1 K_2 : F] \leq [K_1 : F] \cdot [K_2 : F]$$

Moreover, if  $[K_1 : F], [K_2 : F]$  are relatively prime, then the equality hld.

Proof. Let  $\{\alpha_1, \dots, \alpha_n\}$  be a  $F$ -basis for  $K_1$ , and  $\{\beta_1, \dots, \beta_m\}$  be a  $F$ -basis for  $K_2$ .

Then, clearly,  $K_1 K_2 = F(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m)$ .

$$= K_1(\beta_1, \dots, \beta_m)$$

That is,  $K_1 K_2$  is spanned by  $\beta_1, \dots, \beta_m$  over  $K_1$ , hence

$$[K_1 K_2 : K_1] \leq m = [K_2 : F].$$

Now, multiplying  $[K_1 : F]$  to the both side, we obtain the first assertion

And, take  $[K_1 : F] = n$  and  $[K_2 : F] = m$ . Since

$$[K_1 K_2 : F] = [K_1 K_2 : K_2][K_2 : F] \quad (i=1, 2)$$

so  $[K_1 K_2 : F]$  is divided by  $n$  and  $m$ , so  $\ell_{\text{cm}}(n, m) \mid [K_1 K_2 : F]$ .

And, if  $\gcd(n, m) = 1$ , then  $\ell_{\text{cm}}(n, m) = 1$ , so  $nm \leq [K_1 K_2 : F]$ .

Example.

Let  $K = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3}, \sqrt{6})$  and  $K_1 = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3})$  and  $K_2 = \mathbb{Q}(\sqrt{6})$ .

Then,  $K = K_1 K_2$ . And,  $[K_1 : \mathbb{Q}] = 8$ , and  $[K_2 : \mathbb{Q}] = 2$ .

As we expect,  $[K : \mathbb{Q}] = 8 \cdot 2 = 16$ ?

Since  $K_1$  is  $\text{span}_{\mathbb{Q}}\{1, \sqrt[4]{2}, \sqrt{2}, \sqrt[4]{8}, \sqrt{3}, \sqrt[4]{2}\sqrt{3}, \sqrt{6}, \sqrt[4]{8}\sqrt{3}\}$ ,

and  $K_2$  is  $\text{span}_{\mathbb{Q}}\{1, \sqrt{6}\}$ ,

So  $K_2 \subseteq K_1$ ,  $K_1 K_2$  has degree 8.

